

Pearls AQI Predictor

Serverless Multi-Horizon Air Quality Forecasting for Karachi

End-to-End ML Pipeline Report

Muhammad Abdullah Khan

September 2026

Abstract

Pearls AQI Predictor is a serverless machine learning system that forecasts Karachi's US Air Quality Index (AQI) up to 72 hours ahead. It pulls hourly weather and pollutant data from the Open-Meteo API, engineers leakage-free features, and trains a dedicated model for each forecast horizon (XGBoost, Random Forest, and LSTM were compared). Predictions are served through a FastAPI backend and a React dashboard, and the whole cycle – data collection, retraining, and deployment – is automated with GitHub Actions. The best model, XGBoost, is highly accurate in the near term (RMSE 0.92 at +1h, $R^2 = 0.995$) and remains useful a full day out (RMSE 9.41 at +24h, $R^2 = 0.511$). SHAP analysis across all six horizons shows predictions shifting from short-term persistence to weather-driven forecasting as the horizon grows, while confirming the pipeline has no target leakage.

Contents

1	Introduction	3
2	Data Sources and Collection	3
2.1	Open-Meteo API	3
2.2	Location and Historical Data	3
3	Exploratory Data Analysis	3
3.1	Target Distribution	3
3.2	Daily and Seasonal Patterns	4
3.3	Autocorrelation	5
4	Feature Engineering	6
5	Direct Multi-Horizon Modeling	7
5.1	Why Six Separate Models	7
5.2	Model Candidates and Data Split	7
6	Benchmark Results	7

7	Explainability with SHAP	8
7.1	How Feature Importance Shifts With Horizon	8
7.2	Top Drivers at Each Horizon	9
7.3	Directionality	10
8	System Architecture and Deployment	11
8.1	Ingestion	12
8.2	Feature Store	12
8.3	Inference (Backend)	12
8.4	Dashboard (Frontend)	12
9	Automation	12
10	Limitations and Future Work	13
10.1	Current Limitations	13
10.2	Planned Improvements	13
11	Technology Stack	13
12	Conclusion	13

1 Introduction

Air pollution is a serious public-health concern in Karachi, Pakistan's largest city. Pearls AQI Predictor is a complete, serverless pipeline that turns raw weather and pollution readings into 72-hour AQI forecasts:

1. Pull hourly weather and pollutant data from a free public API.
2. Engineer time-aware, leakage-free predictive features.
3. Train a dedicated model for each forecast horizon (1h, 6h, 12h, 24h, 48h, 72h).
4. Serve forecasts through a REST API and an interactive dashboard.
5. Automate the full cycle with GitHub Actions.

The goal: forecasts accurate enough to support health advisories, running entirely on free-tier, serverless infrastructure.

2 Data Sources and Collection

2.1 Open-Meteo API

All data comes from the free, keyless **Open-Meteo API**. It was chosen because it provides weather and air quality from a single source, includes a 3-day weather forecast (essential for direct multi-day prediction), and has generous rate limits for automated pipelines.

Two sets of hourly variables are pulled:

- **Weather** (6): temperature, relative humidity, precipitation, sea-level pressure, wind speed, and wind direction.
- **Air quality** (7): PM_{2.5}, PM₁₀, CO, NO₂, SO₂, O₃, and the US AQI (the prediction target).

2.2 Location and Historical Data

The system targets **Karachi** (24.86°N, 67.01°E), an arid coastal megacity with a mean AQI of about 88; coordinates are configurable for future multi-city expansion. A one-time backfill populated the feature store with three years of hourly history (September 2023 – September 2026, 26,496 rows), processed in overlapping 90-day chunks so that lag and rolling features compute correctly across chunk boundaries.

3 Exploratory Data Analysis

3.1 Target Distribution

Table 1 summarizes Karachi's US AQI over the 3-year dataset. The distribution is right-skewed: most readings fall in the EPA "Moderate" band (51–100), with a tail reaching "Unhealthy" (151–200). Figure 1 shows the distribution and seasonal spread.

Table 1: US AQI descriptive statistics (3-year dataset, 26,496 hourly rows).

Statistic	Value
Mean	88.0
Std	22.0
Min	41
Median	82
Max	173

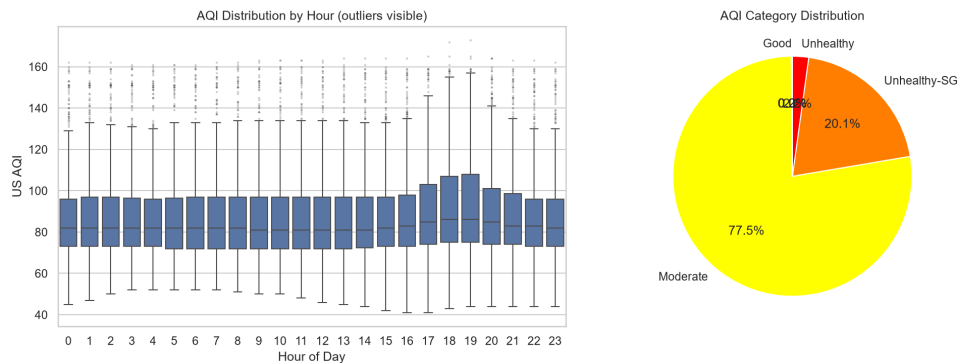


Figure 1: Distribution and outlier analysis for Karachi AQI, showing right-skewed density and seasonal concentration percentiles.

3.2 Daily and Seasonal Patterns

AQI follows two clear cycles, shown in Figures 2 and 3:

- **Daily:** pollution peaks in the evening (7–10 PM) as rush-hour emissions get trapped by the collapsing boundary layer at sunset, and is lowest just before sunrise.
- **Seasonal:** winter (November–February) is worse and more volatile, driven by thermal inversions and calm winds; the summer monsoon (July–September) ventilates the city and washes out particulates.

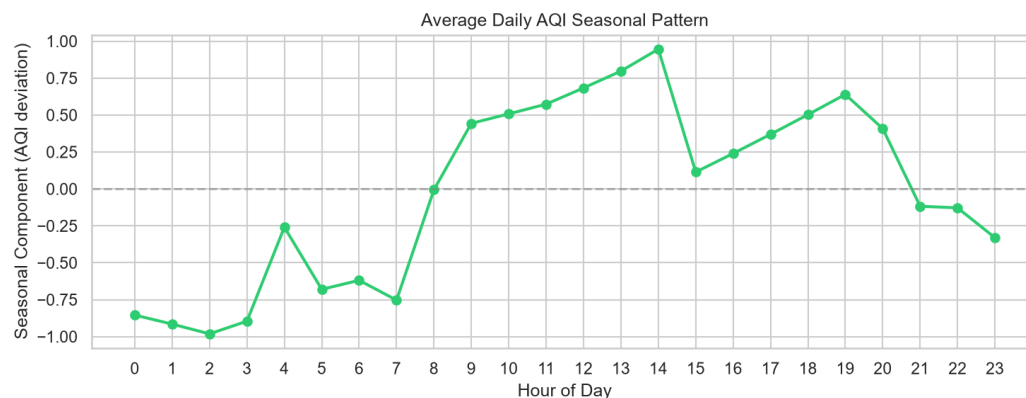


Figure 2: Diurnal hourly AQI profile for Karachi, highlighting morning and evening rush-hour spikes.



Figure 3: Seasonal-Trend decomposition (STL) of hourly AQI into trend, seasonal cycle, and residual components.

3.3 Autocorrelation

An Augmented Dickey-Fuller test confirms the series is stationary ($p \ll 0.001$). The ACF/PACF plots (Figure 4) show strong persistence – autocorrelation is still 0.44 at a 72-hour lag – with sharp PACF spikes at exactly 24h, 48h, and 72h. This is why the model explicitly uses lag and rolling features at those intervals.

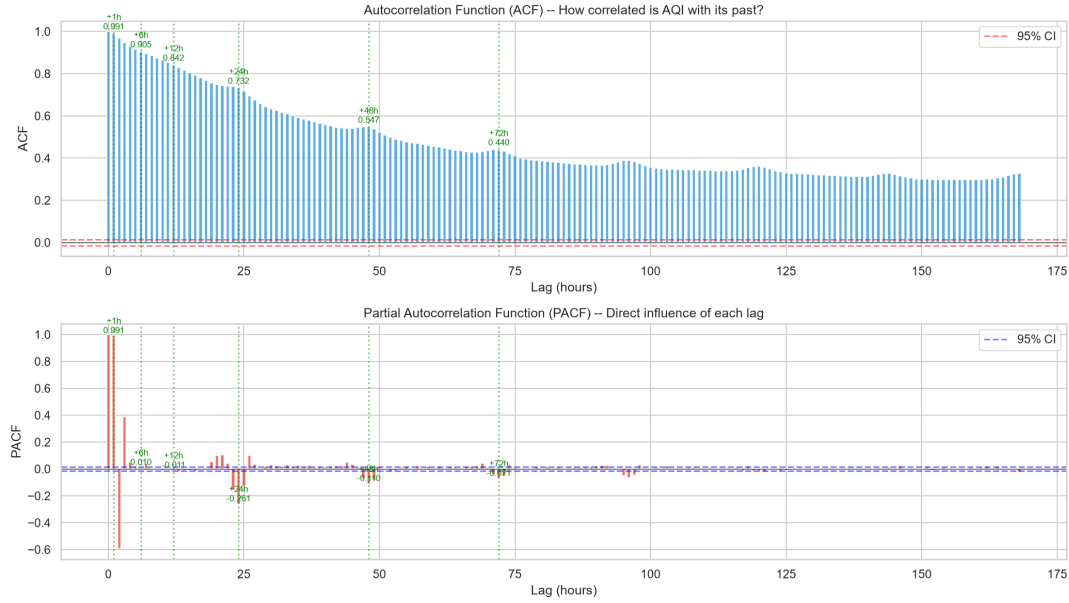


Figure 4: ACF and PACF of Karachi hourly AQI across 72 lags, showing persistence and diurnal spikes at multiples of 24 hours.

4 Feature Engineering

`build_features.py` turns 13 raw variables into 60+ leakage-free predictive features, grouped into six categories (Table 2).

Table 2: Feature categories engineered from raw weather and pollutant data.

Category	Description
Cyclical time	Hour, month, and day-of-week encoded as sine/cosine pairs so boundaries (e.g. 23:00 $\rightarrow$ 00:00) stay continuous, plus a weekend flag.
AQI lags	Past AQI values at $t-1, 3, 6, 12, 24, 48, 72$ hours, capturing multi-day diurnal persistence.
Rolling statistics	Rolling means, standard deviation, min/max, and an exponentially weighted average of AQI (3h–48h windows), all shifted by 1 hour to avoid leakage.
Pollutant history	Only <i>historical</i> pollutant lags and rolling averages are kept; current-timestep readings are dropped entirely (see below).
Trend & momentum	First differences (Δ AQI at 1h, 3h, 6h, 24h) and deviation from rolling baselines.
Weather physics	Wind decomposed into u/v vectors, a temperature $\times$ humidity aerosol-formation proxy, and rainfall/wash-out indicators.

Preventing target leakage. The US AQI is a deterministic function of instantaneous pollutant concentrations, so feeding those same concentrations into the model would let it "cheat." All current-timestep pollutant readings are therefore excluded – only their historical lags and rolling averages are used as features.

5 Direct Multi-Horizon Modeling

5.1 Why Six Separate Models

Rather than a single recursive model that feeds its own predictions back in across 72 steps – compounding errors along the way – the system trains one direct model per horizon:

$$\hat{y}_{t+h} = f_h(\mathbf{x}_t, \mathbf{w}_{t+h}), \quad h \in \{1, 6, 12, 24, 48, 72\} \text{ hours} \quad (1)$$

Hours in between (e.g. +2h to +5h) are filled in by linearly interpolating between the two nearest models. Each horizon’s model also receives Open-Meteo’s 3-day *weather forecast* for that future timestamp ($\mathbf{w}_{t+h}$) plus the forecasted change from now ($\Delta\mathbf{w}_h$), and is pruned to the features that matter most at that range: short lags for near-term models, and diurnal/rolling/forecast features for the 24h+ models.

5.2 Model Candidates and Data Split

Three model families were benchmarked at every horizon: **XGBoost** (500 trees, depth 6, learning rate 0.05, early stopping), **Random Forest** (300 trees, depth 15), and a **PyTorch LSTM** (2 layers, 128 hidden units, 48-hour window). Data was split chronologically – 70% train (18,547 rows), 15% validation (3,974 rows), 15% test (3,975 rows) – to preserve temporal order and avoid look-ahead bias.

6 Benchmark Results

Table 3 shows test-set performance for all three models across all six horizons. XGBoost wins at every horizon.

Table 3: Test-set performance comparison. Bold highlights winning models.

Horizon	Model	RMSE	MAE	R^2
+1h	XGBoost	0.92	0.46	0.995
	Random Forest	1.12	0.48	0.993
	LSTM	1.73	0.82	0.984
+6h	XGBoost	3.94	2.03	0.914
	Random Forest	4.74	2.28	0.876
	LSTM	4.89	2.50	0.869
+12h	XGBoost	6.34	3.57	0.778
	Random Forest	6.76	3.78	0.748
	LSTM	7.05	4.09	0.728
+24h	XGBoost	9.41	6.25	0.511
	Random Forest	9.61	6.38	0.490
	LSTM	9.82	6.69	0.472
+48h	XGBoost	12.07	8.29	0.200
	Random Forest	12.28	8.49	0.172
	LSTM	14.26	9.83	-0.136
+72h	XGBoost	12.60	8.90	0.131
	Random Forest	13.25	9.25	0.040
	LSTM	13.00	9.09	0.042

As expected, accuracy declines with horizon length: near-perfect at +1h, still strong through +12h, and useful for next-day planning at +24h. Beyond a day, skill drops off as multi-day air quality becomes dominated by unpredictable local emissions and weather-forecast drift – the RMSE plateau between +48h and +72h shows the models gracefully converging toward the seasonal climatological average rather than degrading further. XGBoost consistently beats naive baselines (persistence, climatology, same-hour-yesterday) at every horizon.

7 Explainability with SHAP

To understand *why* the models predict what they do, SHAP (SHapley Additive exPlanations) values were computed for all six horizon models. An automated staleness check compares each model’s last-modified time against its saved SHAP report and only recomputes when the model has actually changed, keeping this fast in CI.

7.1 How Feature Importance Shifts With Horizon

Figure 5 and Table 4 reveal a clear pattern: short-horizon predictions lean almost entirely on the AQI’s own recent history, while long-horizon predictions lean on forecasted weather and seasonal timing instead.

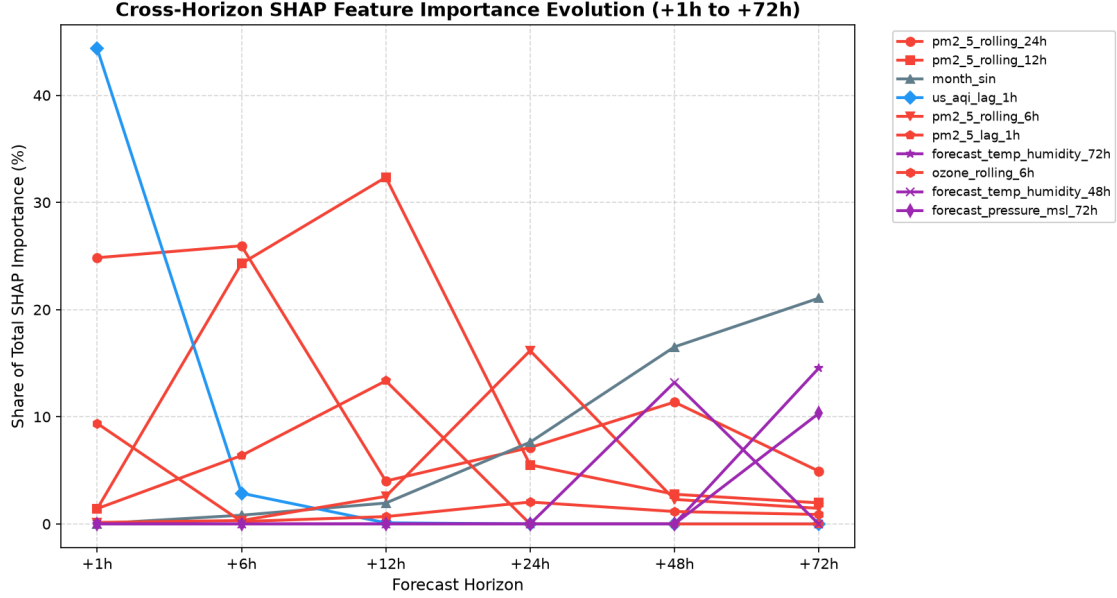


Figure 5: Cross-horizon SHAP feature importance evolution (+1h to +72h), showing the shift from autoregressive lags to forecast weather and seasonal features.

Table 4: Category share of total SHAP importance (%) across forecast horizons.

Feature Category	+1h	+6h	+12h	+24h	+48h	+72h
AQI Lags	37.8%	11.2%	4.7%	4.5%	2.5%	2.0%
Pollutant Lags/Rolling	46.3%	62.4%	62.4%	44.7%	26.6%	15.0%
AQI Rolling Statistics	6.9%	6.2%	3.5%	5.9%	3.6%	3.5%
Trend / Momentum	3.8%	4.8%	2.8%	7.8%	5.2%	4.4%
Current Weather	0.4%	1.5%	4.9%	6.1%	5.0%	4.8%
Forecast Weather / Deltas	0.0%	4.6%	10.8%	21.0%	36.2%	44.1%
Cyclical Time Features	4.8%	9.3%	11.0%	10.1%	20.8%	26.2%

7.2 Top Drivers at Each Horizon

Tables 5, 6, and 7 list the top 5 features by mean $|\text{SHAP}|$ for the immediate, diurnal, and 3-day models; Figure 6 plots them side by side.

Table 5: Top features for the +1h XGBoost model.

Feature	Mean $ \text{SHAP} $
us_aqi_lag_1h	9.854
pm2_5_rolling_24h	5.519
ozone_rolling_6h	2.085
aqi_rolling_3h	1.183
aqi_change_1h	1.044

Table 6: Top features for the +24h XGBoost model.

Feature	Mean SHAP
pm2_5_rolling_6h	5.261
month_sin	2.482
pm2_5_rolling_24h	2.321
forecast_temp_humidity_24h	1.996
pm2_5_rolling_12h	1.795

Table 7: Top features for the +72h XGBoost model.

Feature	Mean SHAP
month_sin	4.616
forecast_temp_humidity_72h	3.186
forecast_pressure_msl_72h	2.260
forecast_wind_u_72h	1.361
pm2_5_rolling_24h	1.080

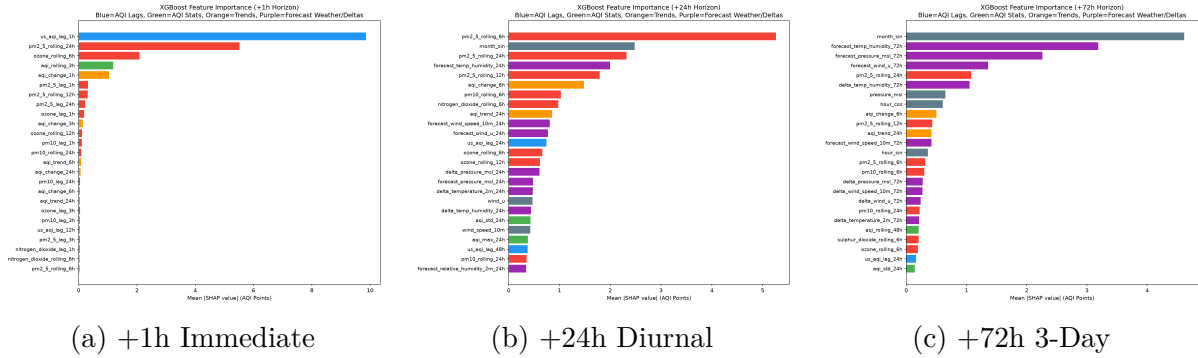


Figure 6: SHAP feature importance across forecast horizons: blue (AQI lags), green (rolling aggregates), orange (trends), purple (forecast weather and deltas).

7.3 Directionality

Figure 7 shows how individual feature values push the +24h prediction up or down: warmer, more humid forecast conditions raise predicted AQI (favoring aerosol formation), stronger onshore wind lowers it (better dispersion), and winter months push the baseline higher.

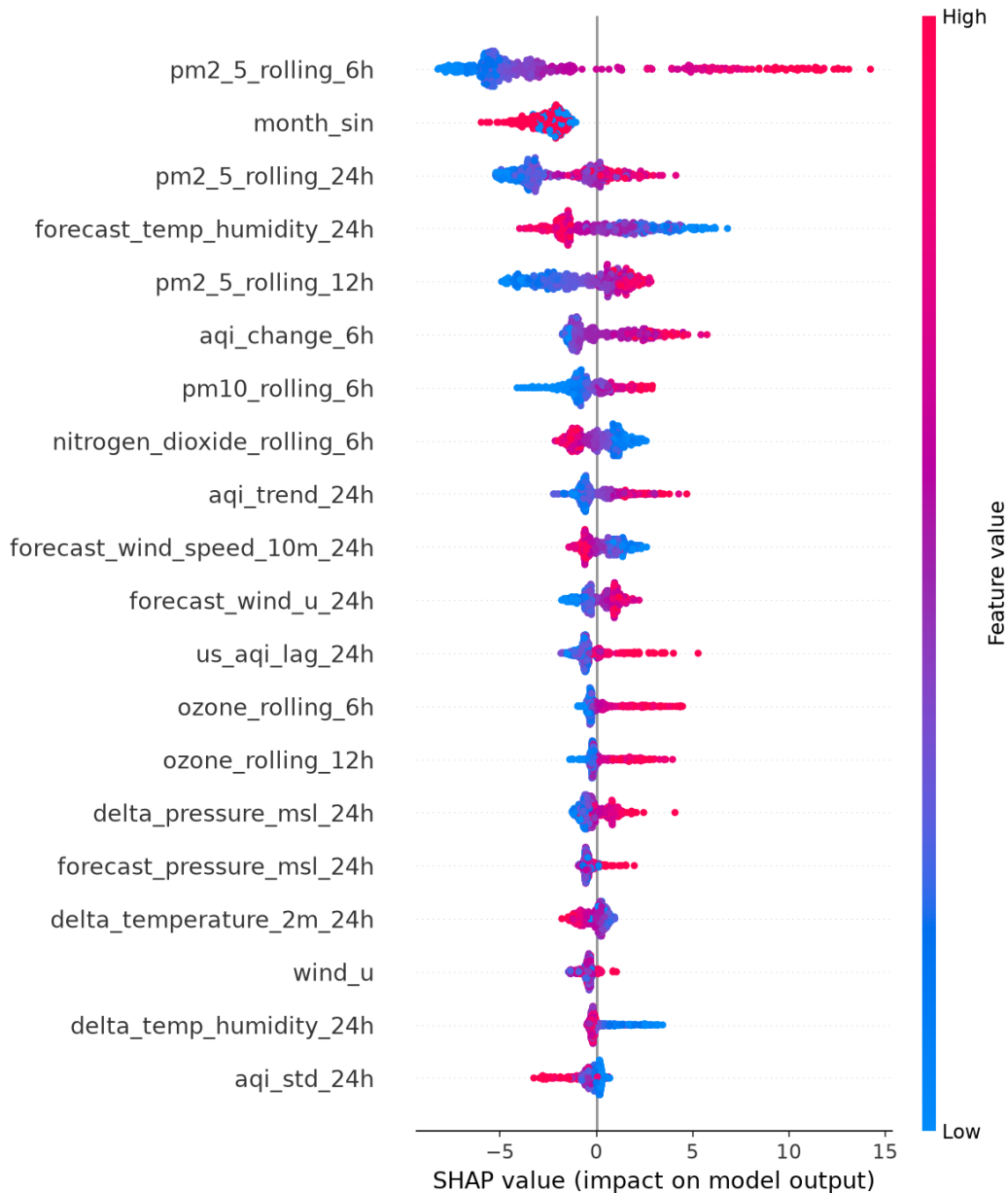


Figure 7: SHAP beeswarm summary for the +24h diurnal model, showing feature value distributions and their positive/negative impact.

Across all six models, no raw (same-timestep) pollutant feature ever appears among the top drivers – confirming the pipeline is genuinely leakage-free, not just leakage-free by design.

8 System Architecture and Deployment

The system is organized into four layers, all running on serverless, free-tier infrastructure with no servers to maintain:

8.1 Ingestion

`sources.py` pulls from Open-Meteo's archive and forecast endpoints and aligns weather and air-quality records on UTC hourly timestamps.

8.2 Feature Store

Engineered features are registered in the **Hopsworks Feature Store** (Apache Hudi format), keyed and timestamped by `time`, which supports ACID upserts, time-travel queries, and reproducible dataset snapshots. `hopsworks_utils.py` manages the connection, schema, and retries on transient network errors.

8.3 Inference (Backend)

A **FastAPI** app (`api.py`) exposes:

- `/api/current` – live AQI, EPA category, and dominant pollutant.
- `/api/forecast` – the full +1h to +72h forecast, interpolated hourly.
- `/api/history` – the trailing 48 hours of observed data.
- `/api/analytics` – model metrics and SHAP importances for all horizons.

It is hosted on **FastAPI Cloud** with automatic deployment on every commit, plus a 5-minute cache to limit calls to Open-Meteo.

8.4 Dashboard (Frontend)

A **React + TypeScript** dashboard (Vite, Tailwind CSS, Recharts) visualizes forecasts and lets users explore SHAP drivers for any horizon. It is hosted on **Vercel**, which proxies API calls to the FastAPI backend.

9 Automation

GitHub Actions drives the whole cycle end-to-end:

- **Hourly feature pipeline** (0 * * * *) – ingests the latest data, recomputes features, and upserts them to Hopsworks in under 2 minutes; a lightweight dependency set keeps this fast.
- **Daily training pipeline** (0 2 * * *) – retrains XGBoost and Random Forest, refreshes metrics, and commits new model artifacts.
- **Continuous deployment** – a new model commit automatically triggers a backend rebuild on FastAPI Cloud.

Together these close the loop: new data comes in, models retrain, and the latest version deploys – all without manual intervention.

10 Limitations and Future Work

10.1 Current Limitations

- **3-day ceiling:** accuracy drops sharply beyond 24 hours (R^2 of 0.13–0.20), since mesoscale air quality depends on local emissions no weather model can predict.
- **Karachi-only:** scaling to other cities will need spatial features or per-city feature groups.
- **Fixed hyperparameters:** tuned once by hand rather than automatically retuned as new data arrives (e.g. via Optuna).

10.2 Planned Improvements

- Satellite aerosol optical depth (AOD) data from Copernicus/CAMS.
- Calibrated uncertainty intervals via conformal prediction.
- Automated alerts when the forecast crosses hazardous EPA thresholds.

11 Technology Stack

Table 8: System technology stack summary.

Subsystem	Technologies
Language	Python 3.12
Machine Learning	XGBoost, scikit-learn, PyTorch
Feature Store	Hopsworks (Apache Hudi)
Model Registry	Hopsworks Model Registry
Data Source	Open-Meteo Weather & Air Quality APIs
Backend Framework	FastAPI, Uvicorn, Pydantic v2
Frontend Stack	React, TypeScript, Vite, Tailwind CSS v4, Recharts
Workflow CI/CD	GitHub Actions
Cloud Hosting	FastAPI Cloud (Backend), Vercel (Frontend)
Explainability / EDA	SHAP, Matplotlib, Seaborn, statsmodels

12 Conclusion

Pearls AQI Predictor shows that a fully serverless, free-tier pipeline can deliver genuinely useful multi-day air-quality forecasts for Karachi – near-perfect one hour out, and still actionable a full day ahead. Pairing horizon-specific gradient-boosted trees with real weather forecasts, rather than one model recursively predicting itself forward, avoids compounding errors and lets each model focus on what matters most at its range. SHAP confirms the story makes physical sense: predictions lean on recent AQI history in the short term and on forecasted weather and seasonality further out, with no shortcuts anywhere in the pipeline.